

INDIVIDUAL PROJECT (LSDS)

WEEK 2 REPORT

(HDFS)

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1) Objective of Week 2:

The objective of Week 2 was to upload the preprocessed large-scale automobile dataset into HDFS and understand Hadoop storage concepts such as block size and replication factor.

2) HDFS Directory Creation

A dedicated project directory was created in HDFS to store the dataset.

Command Used:

```
hdfs dfs -mkdir /user/maria_dev/car_project
```

Verification:

```
hdfs dfs -ls /user/maria_dev
```

This confirmed that the directory car_project was successfully created.

3) Uploading Dataset to HDFS:

The expanded dataset file car_price_big.csv (152MB) was uploaded into the HDFS directory.

Command Used:

```
hdfs dfs -put car_price_big.csv /user/maria_dev/car_project/
```

Verification:

```
hdfs dfs -ls /user/maria_dev/car_project
```

Output confirmed:

- File Size: ~160,003,084 bytes (~152MB)
- Successfully stored in HDFS

4) HDFS Storage Analysis:

Block Size:

Block size was checked using:

```
hdfs getconf -confKey dfs.blocksize
```

Default block size in HDP Sandbox: 128 MB

Since dataset size $\approx$ 152MB:

- Block 1 $\rightarrow$ 128MB
- Block 2 $\rightarrow$ Remaining ~24MB

5) Importance of HDFS

HDFS provides:

- Distributed storage
- Fault tolerance through replication
- Automatic file splitting into blocks
- Scalability for large datasets
- High-throughput data access

In this project, HDFS successfully stored and managed a large-scale automobile dataset for further processing using MapReduce.

6) Conclusion(Week 2):

The automobile dataset was successfully:

- Uploaded to HDFS
- Verified using HDFS commands
- Analyzed in terms of block size and replication
- Prepared for distributed processing in subsequent weeks

```
Using username "maria_dev".
maria_dev@127.0.0.1's password:
Last login: Wed Jan 21 12:57:23 2026 from 172.18.0.3
[maria_dev@sandbox-hdp ~]$ hdfs dfs -ls /
Found 11 items
drwxrwxrwx - yarn   hadoop           0 2018-06-18 15:18 /app-logs
drwxr-xr-x - hdfs   hdfs           0 2018-06-18 16:13 /apps
drwxr-xr-x - yarn   hadoop           0 2018-06-18 14:52 /ats
drwxr-xr-x - hdfs   hdfs           0 2018-06-18 14:52 /hdp
drwx----- - livy   hdfs           0 2018-06-18 15:11 /livy2-recovery
drwxr-xr-x - mapred hdfs           0 2018-06-18 14:52 /mapred
drwxrwxrwx - mapred hadoop           0 2018-06-18 14:52 /mr-history
drwxr-xr-x - hdfs   hdfs           0 2018-06-18 15:59 /ranger
drwxrwxrwx - spark  hadoop           0 2026-02-26 14:02 /spark2-history
drwxrwxrwx - hdfs   hdfs           0 2018-06-18 16:06 /tmp
drwxr-xr-x - hdfs   hdfs           0 2026-01-19 12:24 /user
[maria_dev@sandbox-hdp ~]$ ^C
[maria_dev@sandbox-hdp ~]$ hdfs dfs -mkdir /user/maria_dev/car_project
[maria_dev@sandbox-hdp ~]$ hdfs dfs -ls /user/maria_dev
Found 4 items
drwx----- - maria_dev hdfs           0 2026-01-19 12:24 /user/maria_dev/.Trash
drwxr-xr-x - maria_dev hdfs           0 2026-02-26 14:03 /user/maria_dev/car_project
drwxr-xr-x - maria_dev hdfs           0 2026-01-20 13:59 /user/maria_dev/input
drwxr-xr-x - maria_dev hdfs           0 2026-01-20 13:57 /user/maria_dev/test
[maria_dev@sandbox-hdp ~]$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 172.18.0.2  netmask 255.255.0.0  broadcast 172.18.255.255
    ether 02:42:ac:12:00:02  txqueuelen 0  (Ethernet)
    RX packets 9192  bytes 2633090 (2.5 MiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 8982  bytes 5349260 (5.1 MiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 124610  bytes 139636330 (133.1 MiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 124610  bytes 139636330 (133.1 MiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

[maria_dev@sandbox-hdp ~]$ ^C
[maria_dev@sandbox-hdp ~]$ ls
car_price_big.csv  data.txt  input.txt  mapper.py  reducer.py  test.txt
[maria_dev@sandbox-hdp ~]$ hdfs dfs -put car_price_big.csv /user/maria_dev/car_project/
[maria_dev@sandbox-hdp ~]$ hdfs dfs -ls /user/maria_dev/car_project
Found 1 items
-rw-r--r--  1 maria_dev hdfs  160003084 2026-02-26 14:09 /user/maria_dev/car_project/car_price_big.csv
```

```
[maria_dev@sandbox-hdp ~]$ ^C
[maria_dev@sandbox-hdp ~]$ ls
car_price_big.csv  data.txt  input.txt  mapper.py  reducer.py  test.txt
[maria_dev@sandbox-hdp ~]$ hdfs dfs -put car_price_big.csv /user/maria_dev/car_project/
[maria_dev@sandbox-hdp ~]$ hdfs dfs -ls /user/maria_dev/car_project
Found 1 items
-rw-r--r--  1 maria_dev hdfs  160003084 2026-02-26 14:09 /user/maria_dev/car_project/car_price_big.csv
[maria_dev@sandbox-hdp ~]$ ^C
[maria_dev@sandbox-hdp ~]$ hdfs getconf -confKey dfs.replication
1
[maria_dev@sandbox-hdp ~]$ hdfs getconf -confKey dfs.blocksize
134217728
[maria_dev@sandbox-hdp ~]$ █
```